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Construction 5.0 and Sustainable Neuro-Responsive Habitats: Integrating the Brain–Computer Interface and Building Information Modeling in Smart Residential Spaces

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Abstract: This study takes a unique approach by investigating the integration of Brain–Computer Interfaces (BCIs) and Building Information Modeling (BIM) within residential architecture. It explores their combined potential to foster neuro-responsive, sustainable environments within the framework of Construction 5.0. The methodological approach involves real-time BCI data and subjective evaluations of occupants’ experiences to elucidate cognitive and emotional states. These data inform BIM-driven alterations that facilitate adaptable, customized, and sustainability-oriented architectural solutions. The results highlight the ability of BCI–BIM integration to create dynamic, occupant-responsive environments that enhance well-being, promote energy efficiency, and minimize environmental impact. The primary contribution of this work is the demonstration of the viability of neuro-responsive architecture, wherein cognitive input from Brain–Computer Interfaces enables real-time modifications to architectural designs. This technique enhances built environments’ flexibility and user-centered quality by integrating occupant preferences and mental states into the design process. Furthermore, integrating BCI and BIM technologies has significant implications for advancing sustainability and facilitating the design of energy-efficient and ecologically responsible residential areas. The study offers practical insights for architects, engineers, and construction professionals, providing a method for implementing BCI–BIM systems to enhance user experience and promote sustainable design practices. The research examines ethical issues concerning privacy, data security, and informed permission, ensuring these technologies adhere to moral and legal requirements. The study underscores the transformational potential of BCI–BIM integration while acknowledging challenges related to data interoperability, integrity, and scalability. As a result, ongoing innovation and rigorous ethical supervision are crucial for effectively implementing these technologies. The findings provide practical insights for architects, engineers, and industry professionals, offering a roadmap for developing intelligent and ethically sound design practices.

Keywords: brain–computer interface (BCI); building information modeling (BIM); sustainable architecture; smart residential spaces; neuro-responsive design



Citation: Almusaed, A.; Yitmen, I.; Almssad, A.; Myhren, J.A. Construction 5.0 and Sustainable Neuro-Responsive Habitats: Integrating the Brain–Computer Interface and Building Information Modeling in Smart Residential Spaces. *Sustainability* **2024**, *16*, 9393. <https://doi.org/10.3390/su16219393>

Academic Editor: Aliakbar Kamari

Received: 26 August 2024

Revised: 9 October 2024

Accepted: 26 October 2024

Published: 29 October 2024



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1. Introduction

1.1. Background

In the framework of Construction 5.0, the creation of neuro-responsive environments signifies a noteworthy intersection of neurology, architectural design, and sustainability. This idea uses cutting-edge technologies to develop settings that dynamically adjust to people’s demands and behaviors. The lack of research on integrating Building Information

Modeling (BIM) and Brain–Computer Interfaces (BCI) into architectural design, particularly in ways that facilitate improved human–environment interactions, is a significant motivation for this study.

Notwithstanding the increasing need to tackle environmental deterioration and the requirement for adaptable, eco-friendly living environments, there is still a significant void in the literature about the pragmatic integration of BCI technology into design approaches. In particular, nothing is known about how BCI might allow for in-the-moment modifications in response to occupants' emotional and cognitive states. This disparity makes it more challenging to create flexible, user-centered settings that support sustainability objectives, which makes it difficult to match architectural design to changing environmental and social requirements.

Construction 5.0's evolution of neuro-responsive settings seeks to close this gap by combining ecological sustainability, neurology, and architectural design. As [1] point out, the rapid societal changes and environmental deterioration create an urgent need for novel solutions. To tackle these obstacles, a multidisciplinary approach is beneficial and necessary. Using cutting-edge technology, Construction 5.0 bridges this gap by attempting to build sustainable, user-centric settings sensitive to ecological circumstances and human needs. This underscores the importance of collaboration across disciplines in addressing complex societal and environmental challenges.

According to this study, human–environment interactions will be much more sustainable and adaptable when BCI and BIM are combined in Construction 5.0. It suggests that using BCI to analyze cognitive and affective processes in real-time might result in the development of neuro-responsive environments that adapt to the demands of their inhabitants, improving human well-being and environmental sustainability. This claim is reinforced by [2], who highlight the largely untapped potential of BCI technology in architectural design—mainly in creating personalized and flexible spaces. The potential of BCI technology in architectural design is truly inspiring, opening up a world of possibilities for the future of sustainable living.

The successful integration of BCI and BIM, which can address significant shortcomings in the design methodologies used today, might provide a revolutionary framework for architectural practice in line with the objectives of Construction 5.0. This integrated method, which adapts dynamically to occupants' cognitive feedback, may enhance building performance and help create sustainable, user-centered environments.

This study seeks to close the knowledge gap by examining the synergy between BCI and BIM technologies in residential contexts. Its main objective is to create dynamic, user-responsive spaces that enhance environmental sustainability and occupant well-being. The main goal is to investigate how BCI data may direct the development of personalized, environmentally friendly design solutions.

This study also considers the technological, moral, and legal issues when utilizing cutting-edge technology like BCI and BIM. By tackling these issues, the project intends to contribute to the larger conversation on creating completely integrated, neuro-responsive, and sustainable smart homes.

1.2. Literature Review

This emerging field seeks to redefine how we design, build, and live by utilizing the combined capabilities of Brain–Computer Interface (BCI) technology and Building Information Modeling (BIM), essential components of Construction 5.0. Recent studies by Yang et al. (2023) reveal that the application of BCIs in architectural design remains largely unexplored, particularly during the ideation phase [1,2]. This gap emphasizes the need for customized BCI solutions to enhance the architectural design process. This allows designers to manipulate aspects like window placement and size within virtual environments through their chosen bodily movements. This approach fosters creativity and aligns design with occupants' mental and emotional states, embracing sustainability and human-centric principles.

The relentless pace of urbanization worldwide calls for a shift from traditional development approaches to sustainable and ecological solutions. As Zhang et al. (2018) discuss, such a transformation is crucial for creating healthier, smarter, and more resilient urban futures [3]. These goals are integral to the ethos of Construction 5.0, which focuses on regeneration and sustainability. Bibri & Jagatheesaperumal (2023) describe this innovative paradigm within the context of the Metaverse, a continuous, interactive, three-dimensional network that enhances social interaction and collaboration through advanced technologies [4]. This setup broadens our understanding of technology's potential and propels urban development.

A fundamental aspect of these neuro-responsive habitats is the direct communication enabled by BCIs between the human brain and external devices. Research by Wolpaw, Jonathan R., and Ramadan et al. highlights how this interaction, facilitated through detecting electroencephalographic (EEG) activity, allows control of external devices without physical movement [5,6]. This capability is precious for individuals with motor disabilities. It is crucial to enhance building functionality in the Construction 5.0 framework and integrating BCI with BIM results in dynamic models of residential spaces that can respond to user feedback, significantly reducing energy consumption and waste while minimizing the ecological footprint.

As Sun, Qiuwen, et al. note, EEG's importance in evaluating individuals' psychophysiological states within architectural contexts is significant. They highlight a lack of comprehensive studies in this area, emphasizing the urgent need for further reviews to bridge the gap [7]. This aligns with the rise of the Metaverse and its impact on smart city initiatives, as discussed by Chen, Zefeng, et al. (2024), who recognize the potential of these technological frameworks in enhancing urban living. Additionally, the optimization of HVAC systems by Taheri et al. (2024) demonstrates practical benefits in energy efficiency and thermal comfort, aligning with the goals of Construction 5.0 for efficiency and adaptability [8,9].

Rao et al. (2024) emphasize the importance of adaptable building management systems that respond to changing environmental and occupancy conditions, ensuring stability and efficiency [10]. This resilience is a cornerstone of Construction 5.0, which seeks to make buildings more adaptable and responsive to users' needs. Meanwhile, the neuro-responsive approach, rooted in neuroarchitecture, focuses on creating environments that adapt to human conditions, such as stress or relaxation, enhancing well-being and productivity. This methodology uses insights from neuroscience to develop strategies that promote physical and psychological comfort, as discussed by Xiang, Xin, et al. (2023) [11].

However, achieving fully integrated, sustainable, and neuro-responsive living environments involves significant challenges, including technological and regulatory issues. These include the precision and privacy of BCI implementations, compatibility between BCI and BIM, and the scalability of such advanced solutions. Addressing these challenges requires developing comprehensive regulatory frameworks and standards to keep pace with technological advancements—an essential aspect of Construction 5.0.

Despite these challenges, the potential for neuro-responsive environments to revolutionize residential living is undeniable. As technology advances, the concept of homes that understand and adapt to our needs while promoting environmental sustainability becomes increasingly feasible [12]. Pink et al. (2024) suggest that the evolution of technology and media within homes will foster interactions emphasizing safety, peace, and care, driving a shift towards more integrated, value-driven technological engagements in residential settings [13]. This pioneering approach introduces a new era of residential design characterized by the seamless integration of comfort, interactivity, and sustainability.

Wang et al. (2024) stress the importance of clarifying the vision and foundational elements of this emerging technology, offering insights into its development and successful implementation [14]. Integrating BCI and BIM within innovative residential environments fosters dynamic interactions between inhabitants and their surroundings. This synergy enhances user engagement, optimizes resource management, and advances sustainability efforts, ultimately improving living quality. The ongoing pursuit of these intelligent and

responsive habitats presents a compelling path toward achieving environmental sustainability, exemplifying the innovative spirit of Construction 5.0.

2. Method and Materials

2.1. Ethical Considerations and Privacy

In advanced residential design, integrating Brain–Computer Interface (BCI) and Building Information Modeling (BIM) technologies demands strict adherence to ethical standards, particularly regarding the privacy of occupants’ neural data. Ensuring the integrity and confidentiality of this sensitive information requires the implementation of robust privacy protocols and security measures. These mechanisms are essential to aligning the processes of brain data collection, storage, and usage with ethical guidelines, thereby protecting individuals from potential misuse or unauthorized access.

This process includes comprehensive methods for collecting, processing, and storing sensitive information, which aim to protect individuals’ privacy rights and ensure the security of their data. Alongside these technical measures, deploying BCI systems necessitates a strong emphasis on ethical standards. This approach underscores the importance of user autonomy and informed consent, ensuring that individuals fully understand how their data will be managed and used.

Integrating BCI and BIM technologies marks a paradigm shift in architectural design. This innovative approach aspires to create living environments that are both environmentally sustainable and energy-efficient and profoundly responsive to their inhabitants’ mental and emotional states. By focusing on the seamless interaction between technology and human experience, architects and developers can design residential spaces that significantly enhance occupant well-being.

Such adaptive, human-centric living spaces represent the next frontier in residential design. Achieving this vision requires embedding user experience, privacy, and ethical considerations at the heart of technological implementation. This holistic approach ensures that the development of advanced residential environments remains responsive to user needs and aligned with high moral standards, paving the way for a new era in sustainable, intelligent living.

2.2. Comprehensive Survey of Expert Opinions on BCI and BIM Integration in Residential Environments

The article surveyed 130 participants, selected as experts in research area Z. These participants provided valuable feedback on their experiences and attitudes toward incorporating BCI and BIM technologies in residential environments—the survey aimed to gather diverse responses, ensuring a representative sample and reliable results.

Participants’ feedback was collected using a Likert-scale questionnaire with five response categories, ranging from “1—Strongly Disagree” to “5—Strongly Agree” (Supplementary Materials). This approach enabled a quantifiable assessment of attitudes toward specific study aspects. Notably, the survey achieved a 100% response rate, with all distributed questionnaires returned and considered valid for analysis.

The survey of 130 participants provided robust empirical evidence to support the study’s findings. This strong response rate not only enhances the credibility of the research but also underscores the ethical considerations embedded in the study. The unanimous agreement among participants further highlights the potential for BCI and BIM technologies to revolutionize residential spaces, enhancing both well-being and sustainability.

The study design was meticulously developed to offer deep insights into integrating BCI and BIM within the framework of Construction 5.0. A Likert-scale questionnaire quantitatively evaluated participants’ perspectives on neuro-responsive environments, smart residential spaces, and sustainable architecture. The questions specifically targeted the practical application of BCI–BIM integration and its impact on occupant well-being, energy efficiency, and sustainability.

Participants were chosen based on strict criteria, ensuring they were professionals with expertise in relevant fields such as design, engineering, neurology, and sustainability. This selection process focused on those with experience in Construction 5.0, BCI–BIM research or applications, and advancements in intelligent building technology. This targeted approach ensured a highly qualified and diverse respondent pool, which improved the study's generalizability and relevance.

To ensure reliability, the survey instrument underwent multiple validation steps. Initially, a pilot test with a small group of subject-matter experts was conducted to assess the questions' clarity, coherence, and relevance. Feedback from this pilot phase allowed for necessary adjustments, making the instrument more user-friendly and focused on critical constructs.

Additionally, Cronbach's alpha, a statistical measure of internal consistency, was used to confirm that each set of items accurately reflected the survey's underlying constructs. This thorough methodology strengthened the instrument's validity, providing a solid foundation for subsequent data analysis.

2.3. Methodology for Integrating Brain–Computer Interface (BCI) Technology with Building Information Modeling (BIM) in Residential Environments

This paper investigates the integration of Brain–Computer Interface (BCI) technology with Building Information Modeling (BIM) in residential environments. It aims to create living spaces that adapt to their occupants' mental and emotional conditions. The objective is to synchronize architectural practices with the dynamic interactions between individuals and their surroundings, ultimately fostering more sustainable and adaptable living environments.

A central aspect of the study is its focus on ethical considerations and data privacy. Given the sensitive nature of brain data obtained through BCI, the research underscores the need for robust privacy protocols during data collection, analysis, and storage. Ensuring informed consent is critical, with participants fully aware of how their data will be managed and protected. The study adheres strictly to high ethical standards to preserve the integrity and confidentiality of occupants' data.

The concept of Sustainable Neuro-Responsive Environments is introduced and designed to adapt to occupants' psychological and physiological needs. The project aims to optimize environmental factors, such as lighting, temperature, and spatial arrangements, to enhance comfort and well-being, utilizing BCI technology to achieve these adjustments. This approach seeks to create spaces that respond in real-time to the mental and emotional states of the residents.

The comprehensive data collection process involves noninvasive EEG technology to monitor brain activity in real-time. It also incorporates qualitative methods, including surveys, interviews, and ethnographic studies, to gain deeper insights into residents' experiences and interactions with the built environment. This mixed-methods approach ensures a well-rounded understanding of how BCI technology can influence residential design.

Integrating BCI into BIM requires careful planning, particularly in embedding BCI parameters within the design process and developing adaptive technologies that respond to real-time data. A continuous feedback loop is established, ensuring that the system remains responsive to the changing needs of inhabitants. This adaptability is crucial for developing energy-efficient, sustainable, and user-centered living spaces.

Overall, this study explores the potential of combining BCI and BIM to create a new generation of residential environments. Focusing on technological innovation and ethical responsibility, it aims to pave the way for living spaces prioritizing human well-being and environmental sustainability.

3. Sustainable Neuro-Responsive Habitats

Sustainable Neuro-Responsive Habitats represent a groundbreaking approach to home design, specifically tailored to adapt dynamically to their residents' psychological

and physiological needs. These advanced living spaces are designed to interact seamlessly with a location's existing living and natural systems, forming the essential elements of their architectural framework. At the core of this approach lies the concept of regenerative architecture, with a strong focus on adaptability and environmental integration.

This approach to design emphasizes two primary domains. Goessler, T., and Kaluarachchi, Y. (2023) suggest that homes equipped with adaptive and innovative technologies offer clear advantages in spatial efficiency. They prove that such technologies can reduce the need for separate physical areas dedicated to different functions. By incorporating intelligent technology, these spaces can be customized in three dimensions, allowing them to meet the diverse needs of users with increased flexibility and multi-functionality [15].

Sustainable Neuro-Responsive Habitats utilize cutting-edge Brain-Computer Interface (BCI) technology to optimize critical environmental factors, such as lighting, temperature, and spatial layouts. The aim is to enhance comfort, productivity, and overall well-being within these living spaces. According to Mridha et al. (2021), BCI is a multidisciplinary field that integrates advanced signal processing and sensor technologies to connect brain activity with external systems [16]. Despite notable progress in this area, challenges remain, particularly in enhancing BCI's accuracy, ease of access, and integration into daily life. The study underscores the need for continued innovation in signal processing, hardware development, and BCI applications across various domains.

Advancements in AI and IoT present new possibilities for optimizing smart cities' environmental efficiency and performance. These neuro-responsive environments address ecological challenges and occupant needs by incorporating sustainable design principles, such as energy efficiency and renewable resource utilization. As an integral part of Sustainable Neuro-Responsive Habitats, this approach strives for a harmonious balance between human well-being and ecological resilience by proposing an advanced framework for living environments that are attuned to our internal and external states [17].

3.1. Human-Centric Design Theory and Construction 5.0

User-Centered Design (UCD) is a systematic and iterative approach that emphasizes using investigative and generative methods, such as surveys, interviews, and brainstorming, to understand and address user needs [18] thoroughly. This strategy is crucial in developing products and services that deliver intuitive and satisfying user experiences. By focusing on end-users' perspectives and needs throughout the design process, UCD ensures that the resulting solutions are functional and user-friendly [19].

This approach is precious in contemporary design processes, where user satisfaction is crucial to success in competitive markets. Its cyclical and reflective nature characterizes UCD, continuously revisiting user needs and feedback at various stages of the design process [20]. The user and their requirements are central throughout this methodology, guiding each development phase to ensure alignment with user expectations.

In the UCD framework, design teams actively engage with users throughout the design journey, employing various research and design techniques. The main goal is to develop products that are not only practical but also widely accessible. This user-centric focus ensures that the resulting designs provide significant value to users, addressing their needs directly and effectively.

User-Centric Design Theory is a guiding framework within the complex design matrix that includes architectural spaces, living environments, furniture, and contextual settings. This theory prioritizes the needs, preferences, and behaviors of end-users as the primary drivers for design decisions. As Leppla, Lynn, and colleagues (2021) describe, User-Centered Design is structured through a systematic four-stage process that aligns design solutions with users' practical and experiential needs [21].

This structured approach ensures that design methodologies are more inclusive and responsive, catering to diverse user needs and preferences. Figure 1 illustrates a simplified table of the design process for neuro-responsive environments, detailing steps from understanding user needs to evaluating the impact of design interventions on improving

living spaces. The process highlights the importance of continuous user feedback and assessment to refine and optimize design outcomes, ensuring that the final product meets the evolving needs of its users. Figure 1 shows a simplified table of the design process for neuro-responsive environments, outlining steps from understanding user needs to evaluate the effectiveness of design interventions in enhancing living spaces.

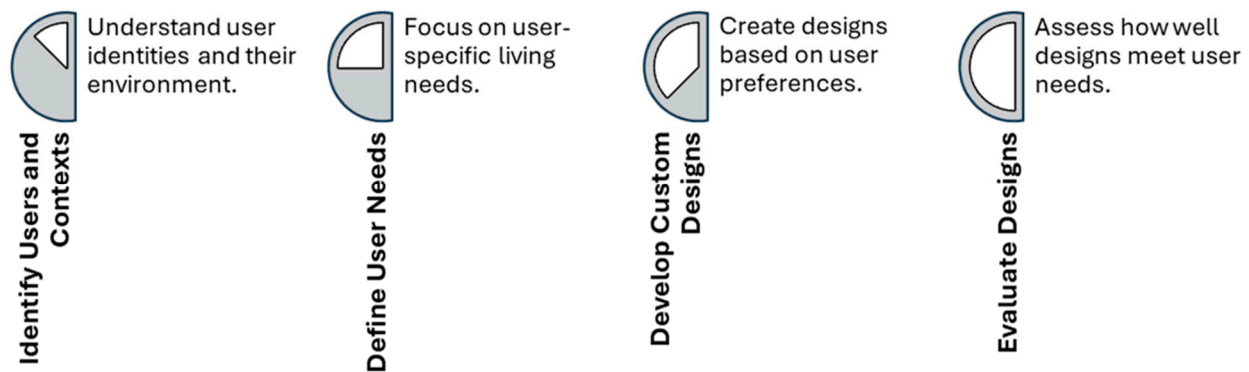


Figure 1. Progressive Shading Patterns of Circular Segments (Source: authors).

User-centric design within the framework of Construction 5.0 emphasizes building solutions that are practical, intuitive, and accessible to users. This approach contrasts with traditional design methods that might prioritize aesthetic appeal or technical innovation without fully considering the end user's needs. The core idea is that effective design must connect with its target users, addressing their unique circumstances, challenges, and goals [22].

The user-centric design process involves thorough research and active engagement with users to gather valuable insights into their experiences and needs. This can be achieved through user interviews, surveys, usability testing, and ethnographic studies. The goal is to develop a deep understanding and emotional connection with users, allowing designers to anticipate and address potential challenges before they become major obstacles.

User-centric design theory emphasizes integrating function and form, recognizing that utility and aesthetic appeal are equally important [23]. A well-designed product should work efficiently and evoke positive emotional responses from users [24]. This integration ensures that design solutions meet practical needs while being enjoyable and satisfying.

Implementing user-centric design leads to developing products, services, and environments that users are more likely to embrace. This acceptance is rooted in the perception that their needs and values are acknowledged throughout the design process. In an era where technology permeates everyday life, this approach ensures that innovations are guided by genuine human needs rather than solely by technological possibilities [25].

User-centric design extends beyond software and consumer products to healthcare services and urban landscapes. It has the potential to enrich the quality, effectiveness, and enjoyment of human interactions in various settings [26]. Centering on the user experience creates more meaningful and engaging interactions between people and their environments, fostering a stronger connection between design and those it serves. Figure 2 depicts the User-Centered Design approach, emphasizing four fundamental components: empathy, accessibility, usability, and iterative design. Empathy guarantees that user insights inform design, whereas accessibility emphasizes the creation of inclusive designs for all users. Usability seeks to deliver a compelling user experience, whereas iterative design entails the ongoing enhancement of the product based on customer feedback. These parts are interrelated, demonstrating the continuous endeavor to enhance the design to satisfy user requirements successfully.

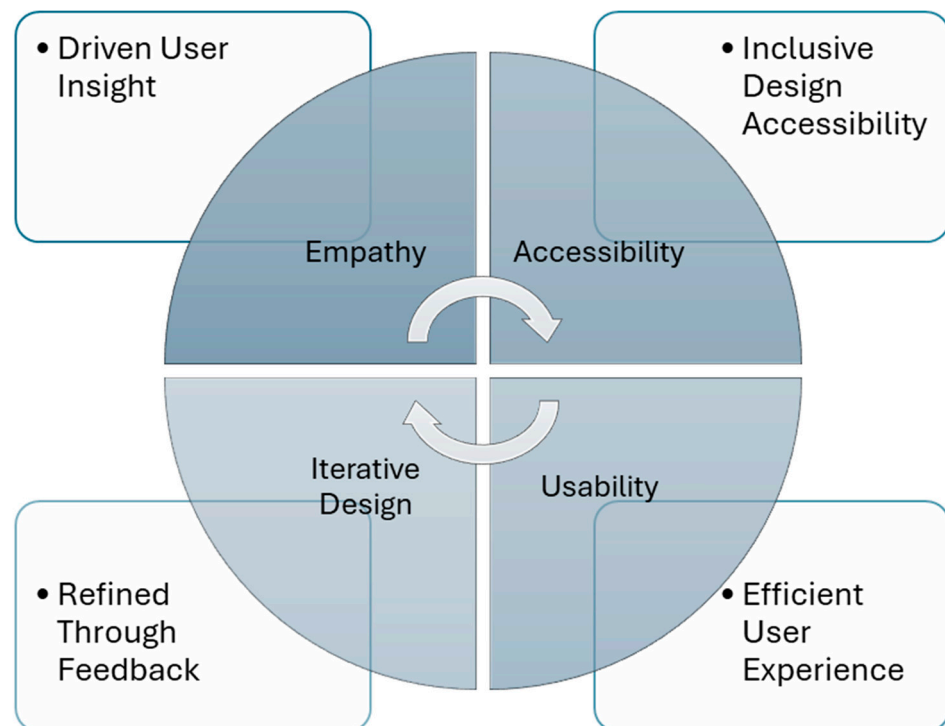


Figure 2. The four pillars of user-centered design (Source: authors).

User-Centered Design (UCD) in architectural design, particularly within the framework of Construction 5.0, involves a structured process with four key phases that are iteratively refined better to align projects with user needs and environmental conditions [27]. This iterative method ensures that architectural projects remain user-centric throughout their development.

The first phase is the analysis, where architectural design teams focus on understanding the complex contexts and environments in which the architecture will operate. During this stage, teams gather empirical knowledge, conducting detailed studies of the space, its users, and their interactions. This foundational research embeds a strong focus on user needs right from the beginning of the design process.

Following the analysis, the specification phase involves clearly defining user needs and requirements. This stage meticulously documents and translates user expectations into concrete design criteria, connecting abstract user desires with practical architectural considerations. The specification phase bridges user feedback and actionable design goals, ensuring that the users' voices are incorporated into every aspect of the project.

The third phase, synthesis, transforms the gathered needs and contextual information into tangible architectural designs. Design teams leverage their creativity and technical expertise to craft aesthetically pleasing solutions that meet user demands. This stage focuses on generating design concepts that balance innovation with practical functionality, ensuring that user requirements remain central.

UCD concludes with the assessment phase [28]. This critical step involves systematically evaluating the proposed design against user needs and the findings from the context analysis. The assessment focuses on usability, accessibility, and environmental compatibility to determine how well the design meets its intended purpose. This phase also helps identify areas where the design might fall short of user expectations or project objectives, guiding necessary revisions.

Each phase of this process iteratively refines the architectural design, ensuring it evolves to suit user needs better and maximize satisfaction and functional performance. The design cycle continues until all user-centric requirements are met, achieving the principles of User-Centered Design [29].

This iterative approach ensures alignment with user needs and adapts the design to changing contexts, resulting in more effective and sustainable architectural solutions. Figure 3 illustrates a schematic depiction of the Brain–Computer Interface (BCI) signal processing pipeline, essential for facilitating interaction between brain activity and external control systems. The workflow consists of three primary stages: Pre-Processing, Feature Extraction, and Signal Recognition. These steps transform unprocessed brain impulses into usable data, allowing for the amalgamation of human cognitive contributions with sophisticated control systems in Construction 5.0 applications. The technology utilizes real-time signal capture to facilitate uninterrupted connections between the brain and machines, enhancing efficiency and responsiveness in construction operations.

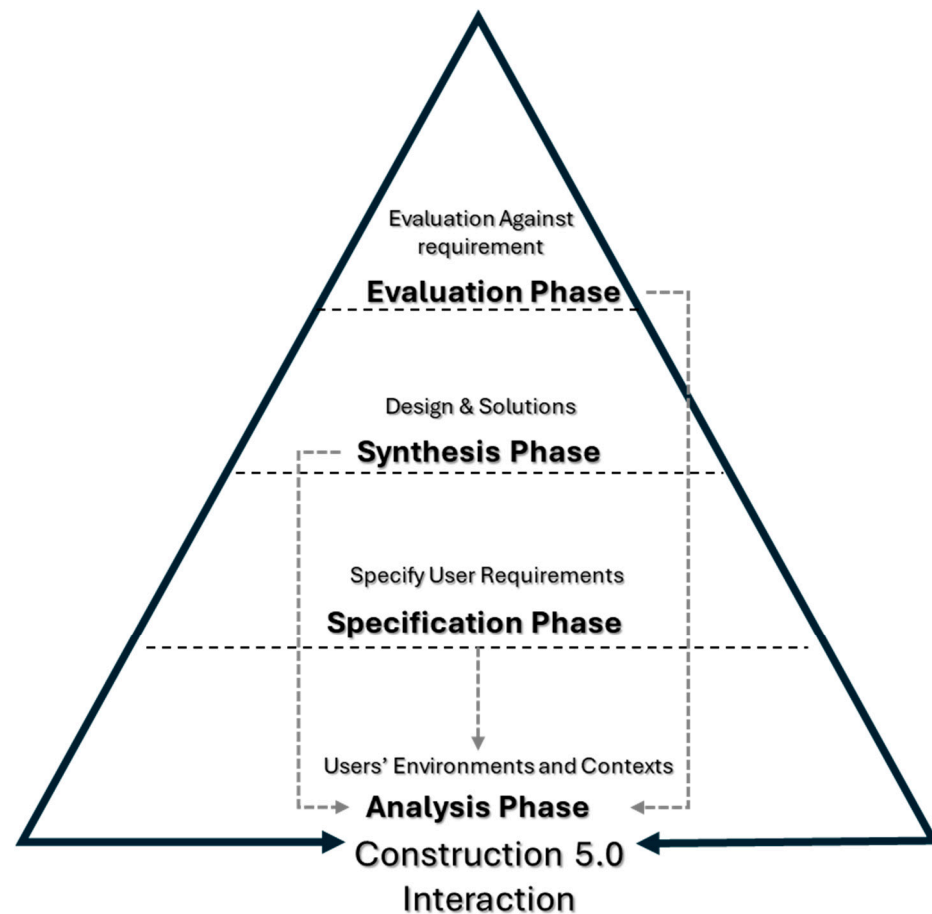


Figure 3. Hierarchical phases of User-Centered Design in architecture.

3.2. Brain–Computer Interface Structure

BCI systems operate on a closed-loop mechanism that provides feedback for user interactions. An imagined hand movement can be converted into a robotic arm command [30]. This seemingly simple act entails a complicated chain of brain functions, one of the body's most sophisticated and vast organs. The brain's billions of neurons and synapses enable complex communication. Figure 4 shows multiple steps in converting brain impulses into actionable commands: Brain activity data are collected and converted into commands to operate virtual and physical applications. The various BCI signal acquisition methods are discussed further.

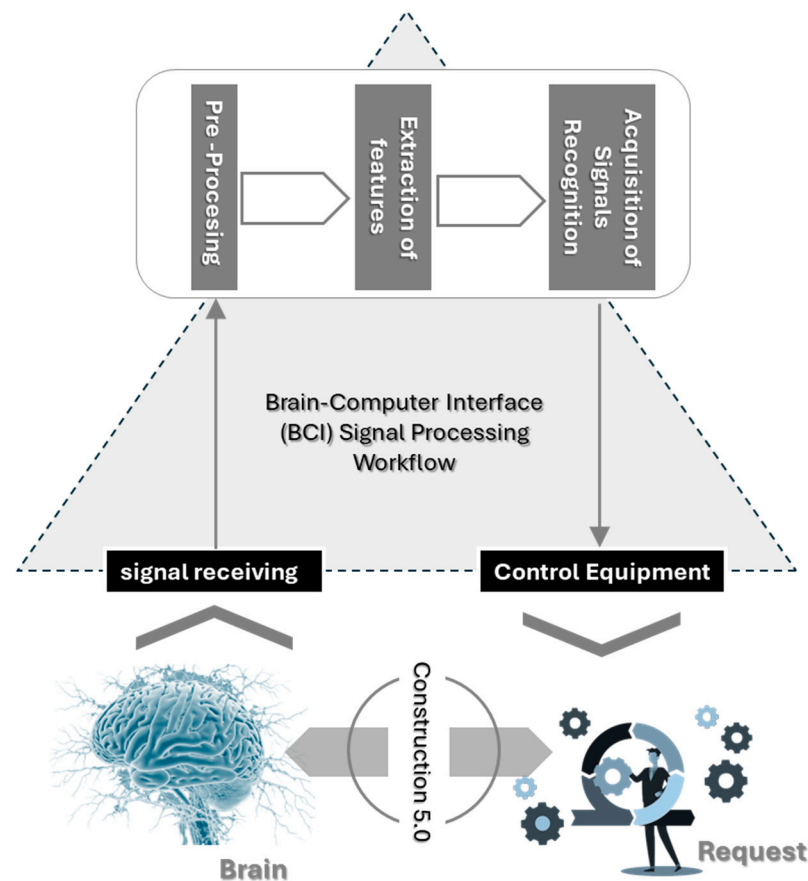


Figure 4. The fundamental structure of a Brain–Computer Interface (BCI) system.

- **Preprocessing:** After signal capture, data must be refined. Brain signals are usually noisy and artifact-filled, requiring filtering and cleaning. This phase is necessary for signal augmentation or signal quality improvement.
- At this point, the focus shifts to signal analysis to extract useful information. Brain activity data are too complicated for simple analysis. Instead, specialized processing algorithms separate key user intent elements.
- **Classification:** The artifact-free signal is classified after feature extraction. This stage is essential for detecting the user’s mental tasks or understanding their orders.
- **Device Control:** Commands the feedback device or application to conclude classification. Classified signals could move a computer cursor or a robotic arm.

As discussed above, the foundational architecture of BCI systems prepares for a deeper look at technological approaches to classifying them [31]. BCI techniques will be detailed in later sections. As seen in Figure 4, the BCI Signal Processing Workflow is illustrated inside the framework of Construction 5.0. The procedure starts with collecting brain signals and then moves to feature extraction and preprocessing. External equipment can be controlled by recognizing and using these properties. Modern technologies like BCI are integrated into smart construction settings, and the workflow shows how brain signals are received, analyzed, and used to control devices. This allows for brain-driven requests to interact with construction equipment.

3.3. Biophilic Design Theory

According to the Biophilic Design Theory, incorporating natural elements into built environments can significantly enhance the physiological health, task performance, and mental well-being of those who inhabit these spaces. Untaru, Elena-Nicoleta, and colleagues (2024) emphasize that this theory highlights the positive impacts of such integration on human health and functionality and offers practical strategies for implementing

biophilic design in residential and workspaces [32]. Biophilic design contributes to establishing environmental design principles that foster a deeper connection between humans and nature.

This approach's fundamental assumption is that humans desire to be close to natural environments. Substantial evidence supports the idea that spaces evoking natural landscapes can significantly enhance the quality of life. This perspective is further validated by Simonsen et al. (2024), who discuss the historical role of nature in aiding healing and rehabilitation, particularly in the context of institutional mental health care [33]. For instance, spaces designed for inpatient treatment have often been crafted to reflect the therapeutic qualities of natural surroundings, underscoring nature's role in promoting well-being.

Neuro-responsive habitats can take this concept further by dynamically adjusting aspects of the interior environment—such as lighting, air quality, and ambient sounds—to better mimic natural conditions. This can be achieved through Brain–Computer Interface (BCI) technology, which allows spaces to respond in real-time to the needs and preferences of their occupants. This symbiotic relationship between advanced technology and biophilic design fosters a more meaningful connection between individuals and the natural world around them, supporting the advancement of sustainability goals [34].

Biophilic design is a fundamental approach to creating architectural spaces that support health, productivity, and psychological satisfaction. It achieves this by replicating natural environments' sensory richness and diversity. This approach advocates for a design philosophy that extends beyond aesthetic considerations, embracing the well-documented benefits of a close human–nature connection. The theoretical framework behind biophilic design promotes a holistic perspective, where the well-being of occupants is considered essential for successful design outcomes [35].

3.4. Feedback Loop Theory in Smart Buildings

Feedback Loop Theory in smart buildings represents a significant milestone in merging architecture with technology, aiming to enhance human living conditions and promote environmental sustainability. The core of this concept lies in a complex network of feedback loops that foster a reciprocal interaction between the built environment and its occupants. This interaction is made possible through the extensive use of sensors, Internet of Things (IoT) devices, and, more recently, Brain–Computer Interface (BCI) technology. These technologies work together to create a dynamic and adaptive living space that responds to and even anticipates the needs of its residents [36].

Applying the Feedback Loop Theory in smart buildings marks a shift toward a more interactive and flexible approach to construction. Sensors and IoT devices play a crucial role by continuously monitoring various conditions, such as temperature, humidity, light levels, and occupant activity. This steady flow of data forms the basis of the feedback loop, enabling the system to make real-time environmental adjustments. These modifications enhance both comfort and energy efficiency, ensuring that the building remains in tune with its occupants' needs.

BCI technology introduces an innovative dimension to these feedback loops. By capturing data related to mental states and physiological responses, BCIs provide a deeper insight into the subjective well-being of individuals [37]. These human-centered data allow buildings to adjust their environments more precisely according to their occupants' specific needs and preferences. For example, the system might alter lighting and temperature based on the time of day, the presence of occupants, or even in response to detected stress levels or cognitive load, thus fostering a space that supports mental and emotional well-being.

Moreover, the Feedback Loop Theory emphasizes the potential of intelligent buildings to contribute to sustainability goals. These systems can significantly reduce a building's environmental footprint by making precise, need-based adjustments. This is achieved by optimizing resource use and minimizing waste, ensuring that energy and other resources are only used when necessary. Such capabilities illustrate the broader impact of smart buildings in advancing environmental sustainability.

Integrating Feedback Loop Theory into smart buildings is a multidisciplinary effort, combining technology, psychology, and environmental science to create spaces that are efficient, sustainable, and highly attuned to human needs [38]. This approach signifies the dawn of a new era for living environments—spaces that are genuinely adaptive to the nuanced requirements of their inhabitants. It opens the door to further innovations that could redefine our relationship with the built environment, paving the way for living spaces that seamlessly integrate human needs with technological capabilities.

4. Integrating BCI and BIM in Smart Residential Spaces

The fusion of Brain–Computer Interface (BCI) technology with Building Information Modeling (BIM) in intelligent buildings represents a significant advancement, enabling environments that adapt to occupants' emotional and cognitive states. This integration allows for the creation of living spaces that respond dynamically to their users' needs, adjusting aspects of the environment like lighting and temperature based on real-time feedback. Despite these advances, challenges such as usability and signal variability continue to limit the broader application of BCIs in real-world settings.

To address these challenges, Simeoni and Ezequiel (2021) propose a methodological framework for improving EEG-BCIs for residential use. Their approach focuses on enhancing classifiers' adaptability and ensuring seamless integration with building automation systems, which is crucial for improving usability and interoperability [39]. This framework is designed to make BCIs more practical for everyday use within intelligent buildings, providing a smoother user experience.

The integration of BCI with BIM combines the precise analysis of brain signals from BCI technology with the detailed environmental and structural information provided by BIM. The process begins with BCI devices capturing and analyzing brain signals to understand occupants' preferences, emotions, and needs in real time. This information is then communicated to the BIM system, which contains a wealth of data on the building's architectural and operational characteristics [40]. This collaboration allows the building to adjust automatically based on the occupants' mental and emotional states.

For example, suppose the BCI detects signs of tension or stress. In that case, the BIM system can respond by adjusting the lighting to a calming spectrum, playing soothing music, or modifying the ambient temperature to enhance comfort. This level of responsiveness improves the occupant experience and contributes to energy efficiency by ensuring that adjustments are made in the most sustainable manner possible, thereby reducing unnecessary energy use.

However, the main challenge lies in developing algorithms capable of effectively processing the complex data generated by BCI systems and integrating these insights seamlessly with BIM to enable real-time adjustments [41]. The ability to analyze intricate brain signals and translate them into actionable changes in the building's environment is essential for the practical application of this technology.

Despite these challenges, the potential benefits are substantial. The promise of customized comfort, improved sustainability, and highly adaptive living spaces makes this integration an exciting development area within intelligent building technology. As research progresses, the vision of homes that can understand and adjust to the needs of their residents in real-time becomes increasingly attainable, opening up new possibilities for the future of smart living.

4.1. Initial Planning and Design Phase

4.1.1. BIM Utilization

Utilizing building information modeling (BIM) during the early planning and design phase is crucial for establishing the foundation of intelligent residential spaces [42]. BIM provides a comprehensive framework for designing both the structural and functional elements of homes, ensuring they are compatible with modern technologies like Brain–

Computer Interface (BCI) from the outset. This early integration helps create structurally sound spaces optimized for the seamless incorporation of BCI technology.

The focus on BCI during the BIM development phase allows designers to integrate necessary parameters directly into the architectural and operational plans. This ensures that the home's design is safe and functional and ready to accommodate advanced technological interfaces like BCI. By including these considerations from the start, the design process aligns more closely with the needs of smart home technologies, facilitating smoother future implementation.

In addition, BIM employs advanced simulation and modeling tools to predict various environmental conditions and potential occupancy scenarios [43]. This proactive approach enables designers to foresee how a smart home might perform under different conditions, allowing them to adjust during the design phase. Such capabilities contribute to the creation of adaptable living spaces that can respond effectively and swiftly to residents' needs, ensuring that comfort and efficiency are built into the structure from the beginning.

This methodology ensures that the design of smart homes prioritizes user comfort and energy efficiency, adapting to changes and providing a better living experience. Table 1 offers a concise overview of BIM's role during the initial planning and design stages of smart homes, highlighting the challenges faced and the opportunities created by integrating BIM with BCI technologies. The table emphasizes how BIM can serve as a strategic tool in overcoming obstacles and maximizing the potential of smart, adaptive residential environments.

Table 1. Integrating BIM and BCI technologies in smart residential design.

Aspect	Details
Purpose	To prepare bright residential spaces with BIM, incorporating BCI from the start.
Framework Development	We are designing homes' structures and functions for easy BCI integration.
Simulation and Modeling	Simulating different scenarios to ensure spaces are adaptable and responsive.
Benefits	Architecturally sound, efficient spaces tailored to occupants' needs.
Challenges	We are integrating BCI with BIM, simulating real-life conditions accurately, and ensuring data security.
Opportunities	Personalized living experiences, improved sustainability, and innovation in smart home technology.

4.1.2. BCI Integration Planning

BCI Integration Planning focuses on seamlessly incorporating brain–computer interface technology within innovative residential environments. The first step, Interface Selection, involves choosing the appropriate BCI technology, with a preference for non-invasive EEG systems [44]. These are favored for user-friendliness and comfort, making them ideal for everyday residential use. Following this, Functionality Mapping is crucial, as it establishes the direct relationship between BCI inputs and the control of building systems. This includes the automation of lighting, HVAC, and even the adjustment of architectural elements like movable walls or windows, allowing for dynamic adaptation of living spaces in real-time [45]. This planning ensures that smart homes can intuitively respond to occupants' needs, enhancing comfort and living experience.

4.2. Technical Integration and Development

4.2.1. BCI System Development

In developing Brain–Computer Interface (BCI) systems, the technical integration and development phase is inherently multidisciplinary, requiring converging concepts from human–computer interaction, software algorithm development, and hardware engineering. This stage aims to create or adapt BCI hardware capable of capturing brain signals with high precision while minimizing the intrusiveness of the device [46]. Reducing the physical

discomfort experienced by users is crucial, alongside enhancing the sensitivity of brain signal detection, which demands advanced technical solutions.

To achieve this, complex methods are employed to enhance the sensitivity of brain signal-detecting systems while maintaining user comfort. This balance between technical sophistication and user experience is critical for successfully deploying BCI technology. At the same time, software algorithms play a vital role in interpreting these neurological signals, converting them into actionable commands within the system's framework. This requires a deep understanding of machine learning and signal processing techniques, ensuring that the system can adapt to the unique brain patterns of individual users. Such adaptability improves the system's accuracy and responsiveness over time.

Developing user interfaces for BCI systems is equally important because it directly impacts the technology's usability and accessibility [47]. Designing these interfaces with the end-user in mind is essential, emphasizing ease of use and straightforward calibration processes. Using principles of intuitive design helps to create interfaces that users can navigate effortlessly, providing clear feedback and guidance throughout their interaction with the system. This focus on User-Centered Design ensures that the BCI system integrates smoothly into the user's environment, making it more practical for everyday use.

Overall, the technical integration and advancement of BCI systems require a collaborative approach that effectively combines sophisticated technological design with the creation of user-friendly interfaces [48]. The ultimate goal is to develop a seamless and efficient interaction between human brains and computer systems, offering users a comfortable and intuitive way to interact with their surroundings through the power of BCI technology. This integration marks a significant step in making BCI systems more accessible and practical for a wider range of applications.

4.2.2. BIM-BCI Integration

A groundbreaking advancement in creating intelligent and responsive environments is the integration of Brain-Computer Interface (BCI) technology with Building Information Modeling (BIM) systems [49]. This integration involves developing a robust data flow architecture that allows for the seamless exchange of real-time neural data between users and BIM models. This continuous data flow enables buildings to make immediate adjustments based on the occupants' cognitive states, creating a more personalized and adaptable environment.

The real-time interaction between BCI and BIM systems allows instantaneous modifications to the building's environment. These adjustments can enhance both comfort and efficiency by dynamically altering various building characteristics, such as lighting, temperature, and spatial configuration, in direct response to the mental and emotional states of the inhabitants [50]. As a result, the space becomes more aligned with the specific needs of its users, offering a more tailored and user-friendly experience.

Another critical aspect of this integration is the development of advanced control systems within the BIM framework that can interpret and respond to inputs from BCI devices. This capability allows for the automation of building operations, meaning that environmental changes can occur without manual intervention. For example, if the system detects a stressed user, it might automatically adjust the lighting to create a calmer atmosphere. This automation streamlines the process of adapting the building environment, ensuring that it remains in sync with the needs of its occupants.

The integration of BCI and BIM improves both the user experience and the building's energy efficiency by optimizing living and working spaces in response to users' specific requirements. The ability to adjust environmental conditions based on real-time data helps reduce energy waste by ensuring that resources are used only when needed and in a manner that aligns with the occupants' well-being.

This BIM-BCI integration marks the beginning of a new era in intelligent building design. It creates environments that respond appropriately to human thoughts and presence, providing a more interactive and adaptable living space. Such innovations represent a

significant leap toward smarter, more responsive buildings that cater to the nuanced needs of their occupants while promoting energy efficiency and sustainability.

4.3. Implementation and Calibration

Integrating BCI technology into intelligent architectural settings is a complex process that requires meticulous attention to detail. This process involves installation and calibration stages, each crucial in ensuring effective integration [51]. During the installation phase, BCI sensors and interactive interfaces are strategically placed within the defined spatial boundaries of the building. This step also includes ensuring that these components integrate smoothly with the existing operational systems, following guidelines established by the Building Information Modeling (BIM) protocol. This integration aims to enable real-time communication between the occupants' brain feedback and the building's responsive systems.

Once the installation is complete, the next step involves calibration and testing. This phase is essential for tailoring the BCI system to match each tenant's unique neurological patterns and preferences [52]. Calibration is a detailed process requiring systematic adjustments to ensure that the system's responses accurately reflect the tenants' neurological inputs. These adjustments fine-tune the system, enhancing its ability to respond accurately to various brain signals and making the interaction between the occupant and the building as seamless as possible.

Extensive testing methods are developed to validate the system's effectiveness. These methods ensure that the BCI accurately interprets neural feedback and that such inputs lead to appropriate changes in the building's environment. The protocols used in this phase are rigorous, reflecting the need for high precision in how the system translates brain signals into environmental adjustments. This ensures that the BCI-BIM integration functions correctly and consistently, providing reliable responses to the users' needs.

This comprehensive approach to installation, calibration, and testing ensures the operational integrity of the BCI-BIM integration. It enhances the building's ability to adapt to the nuanced preferences of its occupants, thus optimizing comfort and efficiency. The integrated system can deliver a highly responsive and tailored living experience by utilizing advanced environmental control mechanisms. This maximizes the comfort of building occupants and improves the overall efficiency of building operations, aligning with the goals of smart and adaptive architecture.

4.4. User Experience and Feedback Loop

The integration of a Brain-Computer Interface (BCI) and Building Information Modeling (BIM) creates a complex system designed to support a highly adaptable and personalized built environment [53]. This mechanism, known as the User Experience and Feedback Loop, operates on the principle of real-time adaptation. It utilizes brain feedback from occupants to initiate immediate adjustments to environmental factors, such as lighting, temperature, and spatial configurations. This process ensures that the environment can dynamically respond to its users' physiological and psychological conditions, aiming to minimize discomfort and improve the efficiency of building operations.

This system accompanies its real-time adjustments with a systematic approach to feedback collection. To achieve this, the system continuously gathers qualitative and quantitative data related to user satisfaction and the overall performance of the environment. This data collection is crucial for the iterative refinement of the BCI-BIM integration, enabling ongoing improvements in the system's accuracy, responsiveness, and overall user engagement. By incorporating this feedback loop into the design, the built environment can evolve, adapting more effectively to the needs and preferences of its occupants.

Including this adaptive feedback loop highlights the importance of User-Centered Design and adaptive learning in developing intelligent environmental systems. It underscores how smart environments can be designed to respond directly to the users' needs, creating spaces that are more comfortable, efficient, and user-friendly. This approach aligns with

broader goals in architecture and technology, focusing on creating environments that can learn from and adapt to their users.

Table 2 presents a scholarly analysis of key challenges and proposed solutions from a comprehensive literature review to explore these ideas further. The table covers Sustainable Neuro-Responsive Habitats, BCI Applications in Architecture and Design, BIM Integration in Smart Residential Spaces, User-Centric Design for Enhanced Living Environments, and Technological Advancements in Smart Habitats. This compilation highlights the progress in developing smart living spaces by integrating innovative technologies and User-Centered Design techniques. It aligns with research findings across multiple disciplines, emphasizing the potential of these integrated systems to redefine the future of residential and adaptive living environments.

Table 2. An extensive analysis of interdisciplinary research on sustainable, innovative habitats, encompassing challenges and solutions in design and technology.

Debated Area	Ref. No.	Challenges	Opportunities and Solutions
Sustainable Neuro-Responsive Habitats	[3,6,14,31–34]	Integrating sustainable architectural practices with neuro-responsive design principles presents challenges in aligning environmental sustainability with the neuropsychological well-being of inhabitants. The complexities arise from the need to balance ecological imperatives with human occupants' cognitive and emotional requirements.	Proposed solutions involve interdisciplinary frameworks that incorporate neuroscientific insights into sustainable design strategies. These include applying EEG-based technologies and biophilic design principles to create living spaces that are both environmentally sustainable and conducive to human well-being.
BCI Applications in Architecture and Design	[2,4,5,15,29,43,45]	Incorporating Brain–Computer Interface (BCI) technology into architectural design processes poses technical challenges, including the development of intuitive interfaces and ethical considerations related to user autonomy and privacy. The field grapples with standardizing methodologies for BCI application in architectural contexts.	Solutions encompass developing noninvasive BCI systems tailored for architectural design, providing architects and designers with novel tools for real-time user feedback and interaction, establishing best practices and ethical guidelines for BCI use, and enhancing the user's influence on design outcomes.
BIM Integration and Smart Residential Spaces	[39–42]	The challenge lies in ensuring seamless interoperability between Building Information Modeling (BIM) and intelligent residential technologies, safeguarding data privacy, and enhancing system security. These challenges extend to optimizing the use of BIM for energy efficiency and sustainability in smart homes.	The advancement of BIM methodologies focuses on creating efficient, sustainable, and occupant-responsive building designs. This involves leveraging BIM to facilitate the integration of intelligent technologies into residential spaces, promoting more thoughtful planning and construction practices.
User-Centric Design for Enhanced Living Environments	[1,4,16–28]	Academic discourse highlights the difficulty of comprehensively understanding and addressing diverse user needs within living environments. This includes overcoming barriers and effectively implementing user-centric design principles prioritizing user satisfaction and accessibility.	The academic response advocates applying user-centric principles and methodologies informed by neuromarketing and organizational behavior insights. This involves a strategic focus on inclusivity, accessibility, and enhancing user experience within designed environments.

Table 2. Cont.

Debated Area	Ref. No.	Challenges	Opportunities and Solutions
Technological Advancements in Smart Habitats	[7–13,30,35–38,44,46–52]	The academic challenge centers on integrating advanced technologies into innovative habitats, such as artificial intelligence (AI), the Internet of Things (IoT), and BCI. This includes addressing scalability, cost-effectiveness, and user adoption of these technologies within urban and residential contexts.	Proposed academic solutions involve developing cutting-edge infrastructure and operational systems for smart city applications. This encompasses leveraging AI, IoT, and BCI technologies to create efficient, responsive, sustainable habitats, emphasizing innovation and operational efficiency.

The connection between Sections 3 and 4 and the empirical results needs to be clarified to address the concerns raised by the reviewer. Section 3, titled “Sustainable Neuro-Responsive Habitats”, presents the theoretical foundations and conceptual frameworks that guide the research. This chapter delves into advanced architectural techniques like User-Centered Design (UCD) and Brain–Computer Interface (BCI) technology. These technologies provide the basis for understanding how adaptive systems can be integrated into neuro-responsive and environmentally sustainable environments. This section establishes the critical background necessary to contextualize the innovative design approaches explored throughout the research.

After laying this theoretical groundwork, Section 4 transitions into a technical analysis of how Building Information Modeling (BIM) and Building Control Interfaces (BCIs) can be incorporated into smart residential environments. This chapter focuses on these technologies’ technical aspects and potential, exploring their role in creating more adaptive and responsive living spaces. Integrating these frameworks provides insight into how architectural design and technology can work together to optimize living environments.

The empirical findings, derived from a survey of 130 participants, are not directly presented within these theoretical and technical discussions in Sections 3 and 4. Instead, the survey data are analyzed in subsequent sections, which are directly related to the proposed models. The survey results evaluate the real-world application of the frameworks outlined in the earlier chapters, mainly focusing on the adaptability and efficiency of neuro-responsive settings.

Sections 3 and 4 thus lay the foundation by offering a comprehensive overview of the theoretical and technological frameworks. The data collected from the survey allow for an assessment of how these frameworks perform when implemented in real-world applications. This empirical analysis evaluates the practical viability and effectiveness of the proposed neuro-responsive environments.

The discussions in Sections 3 and 4 serve as a crucial foundation, leading to the empirical application and analysis presented in later chapters. This structure ensures that the theoretical and technical context is well-defined before introducing empirical evidence, providing a more apparent distinction between the background information and the survey-based findings. This clarification aims to better differentiate the theoretical discussions from the empirical analysis, offering a more cohesive understanding of the research flow and outcomes.

5. Results and Discussion

The participants’ responses were analyzed using a Likert-scale survey with five response categories ranging from “1—Strongly Disagree” to “5—Strongly Agree”. This approach allowed researchers to quantify attitudes towards specific statements, capturing the range of agreement among the study participants. The survey included 130 participants, achieving a 100% response rate, as all distributed questionnaires were returned and considered valid for analysis.

The majority of respondents (55.4%) selected “4”, indicating a high level of agreement, though not the maximum possible level. This suggests that while many participants strongly agreed with the statement, there was a preference for expressing agreement just below the highest level. Following this, the “5—Strongly Agree” category accounted for 25.4% of responses, reinforcing the overall positive sentiment towards the statement.

Responses indicating neutrality or disagreement were significantly less common. 17.7% of participants chose the “3” (neutral) response, reflecting a smaller segment that neither agreed nor disagreed strongly with the statement. Even fewer participants selected “2” (disagree), with only 1.5% indicating a low level of agreement. Notably, no participants selected “1—Strongly Disagree”, suggesting a lack of strong opposition among the surveyed group.

The statistical analysis of the responses further supports this trend toward agreement. The mean response value was calculated at 4.05, with a median value of 4, highlighting the tendency of reactions towards the positive end of the scale. The mode—the most frequently selected response—was also “4”, aligning with the predominant agreement observed in the survey.

This methodological approach provided a clear and quantifiable understanding of the participants’ collective opinions, emphasizing the consensus towards agreement. The structured nature of the survey data also facilitated further statistical analysis and enabled deeper discussions and interpretations of the group’s perspectives. The results illustrate a robust overall agreement, with minimal divergence, offering valuable insights into the participants’ attitudes.

5.1. BCI Technology and Interaction with Living Spaces

BCI technology transforms people’s interactions with their environments by directly connecting the brain and computer systems. This capability allows users to control various aspects of their surroundings more effortlessly and intuitively. Through BCIs, individuals can directly influence elements of their living space, such as lighting, temperature, and other environmental controls, enhancing their ability to customize their homes to suit their preferences.

The potential of BCI technology is especially significant in designing adaptive living spaces. It allows for environments that respond dynamically to the needs and well-being of their inhabitants. By leveraging the direct brain–device interface, BCI technology makes it possible for living spaces to adapt in real time, creating an atmosphere that can improve overall comfort and support better mental and physical health.

Figure 5 presents detailed survey data on the integration and user acceptance of BCI technology in residential settings. This figure is divided into four charts, each highlighting different aspects of how users perceive and engage with BCI-enhanced environments. The data provide valuable insights into how comfortable people feel using this technology, their enthusiasm for adopting it, and their overall acceptance of BCIs as a tool for managing their home environments.

The information from these charts helps to understand the public’s readiness to embrace BCI technology in their daily lives. It sheds light on the factors influencing user satisfaction and the likelihood of broader adoption. Understanding these aspects is crucial for advancing BCI applications in residential design and ensuring that the technology meets the needs and expectations of its users. Through the analysis of these answers, we may achieve a deeper comprehension of the potential influence of BCI technology on the development of adaptable environments.

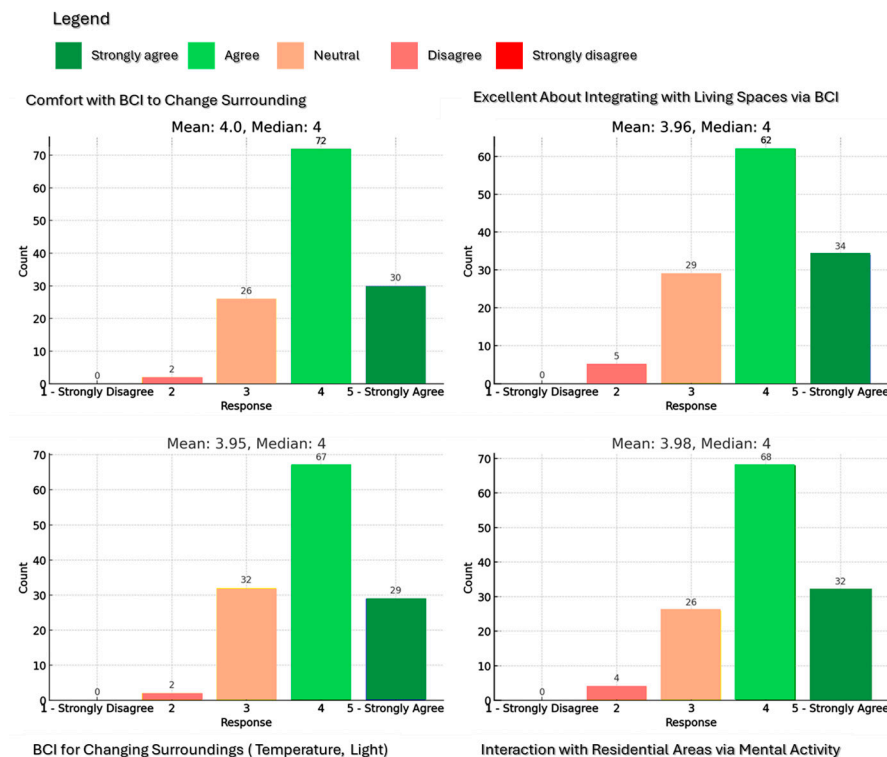


Figure 5. Survey responses on the integration and interaction of (BCI) technology with living spaces (Source authors).

5.1.1. Comfort with BCI to Change Surroundings

The results indicate a significant and favorable reaction to the comfort level in utilizing BCI technology to modify one's environment. Out of 130 participants, most (72) evaluated their comfort level at a score of 4. Additionally, a substantial proportion of 30 people strongly agreed with this rating. This demonstrates a significant degree of receptiveness and eagerness to incorporate BCI technology into everyday life since it can create a customized and adaptable environment that caters to the user's requirements.

5.1.2. Excitement About Interacting with Living Spaces via BCI

Another critical factor is the enthusiasm for interacting with living places through BCIs. The participants exhibited a significant degree of curiosity, with an average score of 3.96. This enthusiasm arises from the promise of BCIs to provide immersive and interactive environments, boosting the quality of life by adjusting to the user's cognitive and emotional conditions.

5.1.3. BCI for Changing Surroundings (Temperature, Light)

BCI technology may manipulate and regulate environmental conditions, including temperature and illumination. According to the poll results, 67 participants gave it a 4, and 29 gave it a 5. This indicates a significant level of comfort. This exemplifies the aspiration for more authority over one's surroundings, which contributes to creating a customized and cozy living area.

5.1.4. Interaction with Residential Areas via Mental Activity

The promise of Brain–Computer Interface (BCI) technology to enable contact with residential settings through mental activity is highly regarded. The average number of 3.98 indicates a high level of curiosity, with several people enthusiastic about exploring this advanced method of engagement. This skill has the potential to significantly influence the way we control and interact with our living spaces, resulting in them being more adaptable and in harmony with our psychological and emotional requirements. Incorporating BCI

technology into living spaces signifies a substantial advancement toward more intelligent and adaptable surroundings. The favorable acceptance of BCI applications in modifying environments, regulating environmental variables, and facilitating interaction via cognitive processes underscores the promise of this technology to improve welfare and provide more individualized living experiences. BCI technology's ongoing development will undeniably impact the future of home automation and smart living.

5.2. Environmental and Eco-Friendly Technologies in the Home

Incorporating environmentally friendly technologies into homes is becoming increasingly important in sustainable living. These technologies help reduce the carbon footprint, enhance energy efficiency, and create healthier homes and communities. Homeowners can significantly reduce their environmental impact by integrating innovations such as solar panels, energy-efficient appliances, smart home systems, and sustainable building materials. These advancements collectively encourage more sustainable living practices and promote the conservation of resources.

The shift toward environmentally friendly housing is driven by growing awareness of environmental challenges and a shared desire to build a more sustainable future for coming generations. This shift is reflected in survey data highlighting the strong appeal and perceived importance of integrating eco-friendly technologies in residential properties. The first figure examines the attractiveness of living in a location with environmentally friendly technologies. It reveals that a significant portion of participants (48.5%) rated the appeal at level 4, with an additional 28.5% assigning the highest rating of 5. This distribution of opinions indicates broad agreement on the appeal of eco-friendly household technologies, supported by an average rating of 4.02 and a median of 4, indicating a general trend toward positive attitudes.

In addition to the appeal of eco-friendly homes, the balance between environmental sustainability and energy efficiency in residential environments is a crucial concern. The second graphic highlights the importance of achieving this balance, showing that 38.5% of participants rated it a 4, while 25.4% gave it a 5. This suggests a strong recognition of the value of sustainable energy management practices in residential settings. The average rating of 3.83 and a median of 4 further reflect a consensus among respondents on the significance of this topic.

Integrating environmentally friendly technology into homes encompasses several key areas. For instance, solar photovoltaic systems provide a renewable source of electricity, reducing reliance on non-renewable energy sources. Energy-efficient appliances and smart home systems help optimize energy use, lowering utility costs and reducing environmental impact. Using sustainable building materials, such as bamboo and recycled steel, supports the conservation of natural resources and helps minimize waste.

However, implementing environmentally friendly technologies in homes is not without challenges. A primary barrier for many homeowners is the high initial cost of installing solar panels and energy-efficient systems. Additionally, there is often a lack of widespread understanding of the long-term economic and environmental benefits these technologies can offer. Integrating such technologies into existing homes can also be complex, requiring significant adjustments and modifications.

Despite these challenges, there is clear evidence of growing acceptance and recognition of the need for eco-friendly technology in residential settings. As awareness increases and technology improves, more homeowners are expected to adopt sustainable living practices. This shift will contribute to a healthier planet and more resilient communities, aligning with global environmental goals and fostering a more sustainable future for residential living.

Figure 6 provides insights into the respondents' agreement on two key aspects of an eco-friendly lifestyle, further emphasizing the positive outlook towards sustainable living. The data underscore the potential for widespread adoption of sustainable practices, offering hope for the future of environmentally conscious residential design and community development. In Figure 6a, 59.2% concur that reconciling eco-friendliness with energy-

efficient technology is significant, with a mean score of 4. Figure 6b indicates that 50% of respondents agree that residences should accommodate eco-friendly lifestyles, yielding a mean score of 4.02.

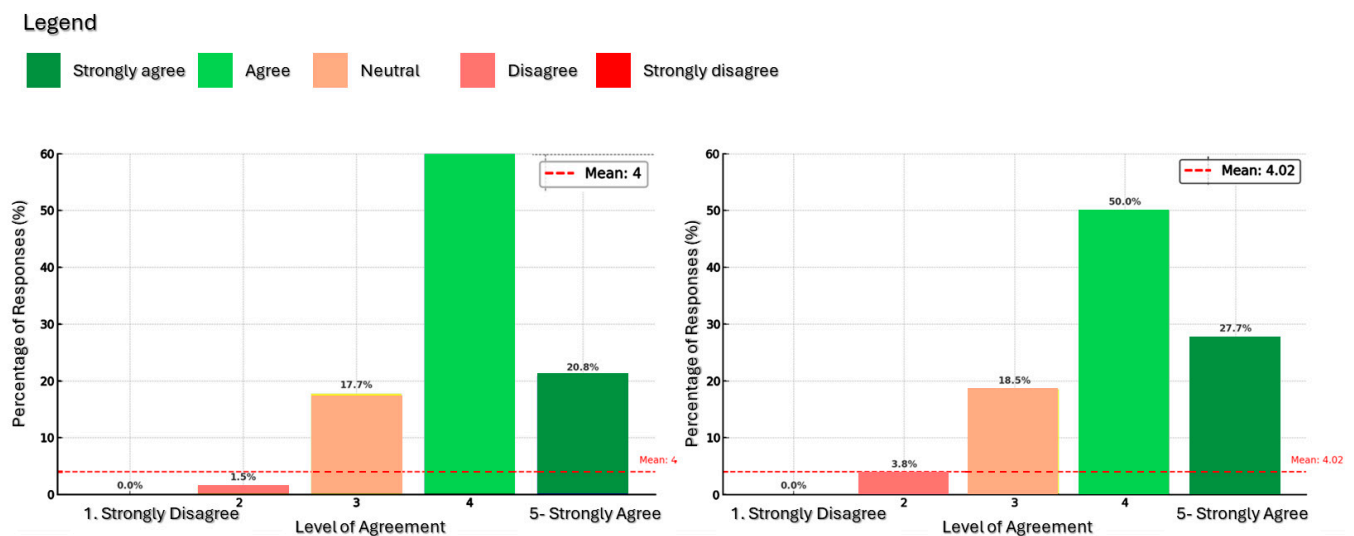


Figure 6. Survey responses on the appeal and importance of eco-friendly home technologies and energy efficiency (Source authors).

5.3. Impact of Living Settings on Well-Being

The living environment plays a crucial role in shaping an individual's health and overall happiness. Natural light, air quality, noise levels, and access to green spaces strongly influence mental and physical well-being. Adaptive and environmentally friendly living environments, which can adjust according to their residents' emotional and mental states, are particularly effective in enhancing comfort and promoting a healthier lifestyle. Integrating technology that monitors and responds to these factors within living spaces can significantly improve quality of life, reducing stress and fostering a sense of well-being.

When technology is embedded into living environments to monitor these conditions, it creates a space that can adapt to the needs of its occupants. This adaptability allows for real-time adjustments, such as altering lighting levels based on mood or adjusting airflow to maintain optimal air quality. As a result, these technologically advanced environments reduce stress and enhance overall life satisfaction, making them a valuable approach to modern housing design.

Building homes prioritizing health and happiness requires a deep understanding of how living environments influence well-being. Designers and architects must consider various environmental factors' psychological and physiological impacts and incorporate solutions that address these needs. This approach ensures that living spaces are not just functional but actively contribute to the well-being of those residing there.

The impact of adaptable living environments on well-being is supported by survey data, as shown in Figure 7. Over half of the survey participants (55.4%) rated the influence of such environments on well-being at a level of 4, while an additional 25.4% rated it at the highest level of 5. This strong consensus suggests widespread recognition of the positive effects of adaptable living spaces on improving residents' quality of life. The average response score of 4.05 further underscores the general agreement on the importance of adaptable living arrangements in promoting well-being.

A small minority of participants rated the impact lower, with only 17.7% choosing a rating of 3 and very few expressing disagreement with ratings at levels 1 or 2. This indicates that most respondents perceive adaptable living environments as having a substantial and beneficial impact on their mental and emotional states.

The findings emphasize the importance of creating environments sensitive to their inhabitants' emotional and mental conditions. The ability to adapt and respond to changes

in mood or stress levels not only enhances comfort but also plays a vital role in fostering a healthier lifestyle. This research highlights the potential of adaptive technologies to transform residential spaces into more responsive, supportive, and health-promoting environments.

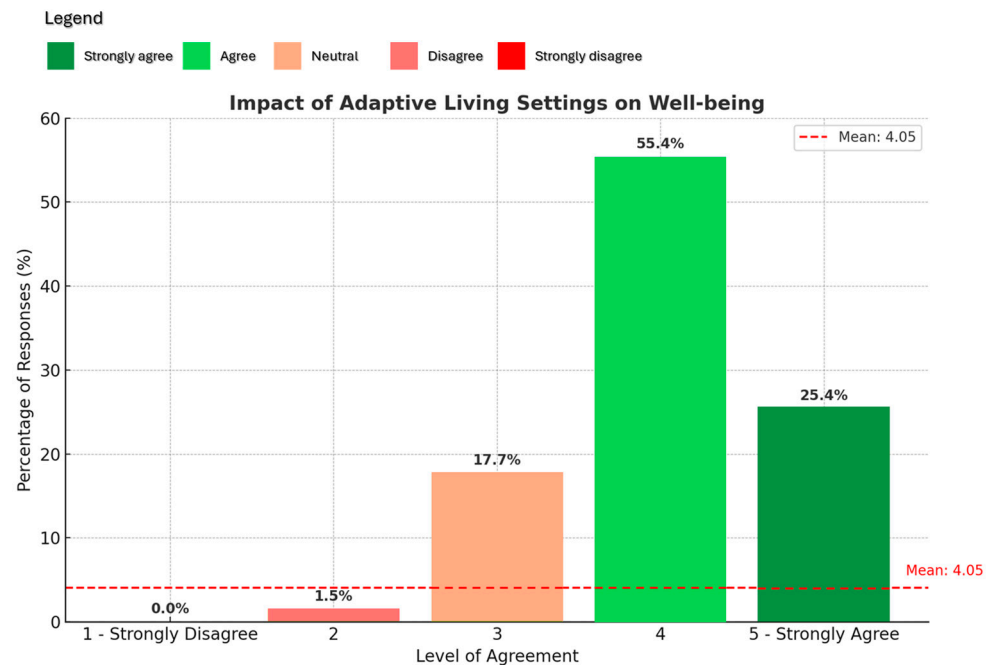


Figure 7. Survey responses on the impact of adaptive living settings on well-being (Source: authors).

5.4. Building Information Modeling (BIM) in Home Planning and Maintenance

Building Information Modeling (BIM) has become a transformative approach in the construction and maintenance industries, especially in residential planning and upkeep. BIM involves creating detailed digital models representing a building's physical and functional characteristics. This enables stakeholders to visualize and manage building data throughout its entire lifecycle. By using BIM, construction projects can achieve higher quality, greater efficiency, and more informed decision-making. Additionally, BIM's digital twin technology provides a comprehensive view of a home's structure and systems, improving planning, execution, and the management of repairs and renovations.

The importance of BIM technology in residential construction and management is highlighted in recent study findings. These findings emphasize the role of digital twin technology in enhancing decision-making processes for home improvements and repairs. Survey results show a strong positive trend towards recognizing the value of digital twins created through BIM. Among the 130 survey respondents, none strongly disagreed with the importance of this approach, and only 1.5% (two respondents) rated it as less significant. A small segment of 17.7% (19 respondents) rated the importance moderately (3), indicating some uncertainty about its impact.

However, most participants strongly supported the significance of BIM in improving decision-making related to home maintenance. Sixty percent of respondents (78 participants) rated its importance as a 4, while 28% (27 participants) gave it the highest score of 5. With a mean value of 4 and a median value of 4, the survey results indicate a consensus on the critical role of digital twin technology in guiding effective home renovation and repair decisions. The survey achieved a 100% response rate, reflecting thorough engagement from all participants.

Similar positive sentiments were found in a related survey focusing on the standardization of BIM technology in residential construction and design. This study aimed to assess the necessity of consistency in BIM practices to maintain quality and ensure the durability of buildings. None of the respondents strongly disagreed with the need for

standardization, and only 3.8% (five respondents) disagreed, rating it a 2. A segment of 18.5% (24 respondents) expressed uncertainty, giving it a score of 3.

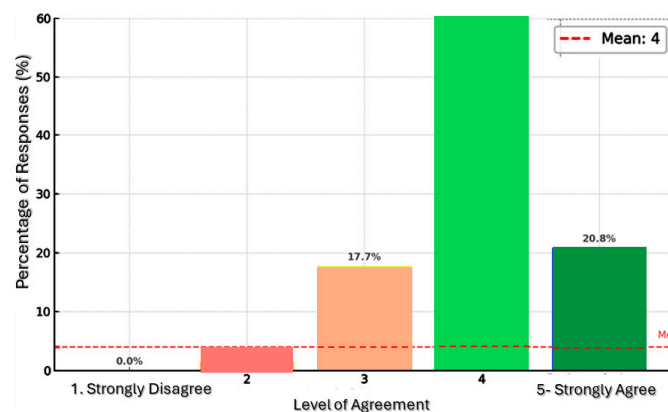
Most respondents supported the need for standardized BIM practices, with 50% (65 participants) agreeing with the statement by scoring it a 4 and 27.7% (36 participants) strongly agreeing, giving it a rating of 5. The results from this survey showed a mean value of 4.02 and a median of 4, indicating widespread recognition of the importance of consistent BIM technology standards in the construction and planning industries. Such standardization is seen as crucial for maintaining quality standards and ensuring the longevity of residential buildings.

Collectively, these findings highlight strong support for adopting digital twin technologies through BIM and for standardizing BIM practices in home construction and renovation. This endorsement reflects a broad professional consensus on the value of these approaches for enhancing the quality of residential environments and ensuring well-informed decision-making throughout the lifecycle of buildings. The positive reception of BIM's capabilities demonstrates its critical role in the future of sustainable and efficient residential construction. Generating a digital twin using Building Information Modeling (BIM) for house repairs and restorations is an important subject. Figure 8 demonstrates the survey's findings on two critical topics: the requirement of consistent BIM technology in home building and the value of generating a digital twin. Both surveys show a high level of agreement, with mean values of 4 and 4.02, respectively, highlighting the variables' perceived significance.

Legend

Strongly agree Agree Neutral Disagree Strongly disagree

Importance of creating a Digital Twins through BIM for Home Repair



Importance of Uniform BIM Technology in Home Building

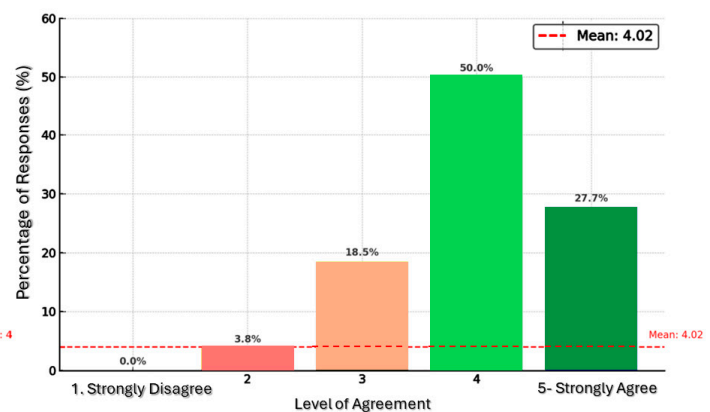


Figure 8. Survey results on the importance of digital twins through BIM for home repairs and renovations and the need for uniform BIM technology in home construction (Source: authors).

Building Information Modeling (BIM) and Brain–Computer Interfaces (BCIs) represent transformative residential design, construction, and management technologies. These innovations align with the principles of Construction 5.0, which emphasizes human-centered design, technological advancement, and sustainability. The study comprehensively assesses attitudes toward these technologies, highlighting the strong consensus on their potential benefits.

The survey results show broad support for using BIM in creating digital twins to improve decision-making in home repairs and renovations. With a mean score of 4, respondents demonstrated an understanding of the value of accurate digital representations of buildings' physical and functional characteristics. This technology enhances decision quality, boosts project efficiency, and ensures a higher quality standard. Furthermore, the

need for consistency in BIM practices in residential buildings was emphasized, suggesting that standardized processes are essential for maintaining quality and durability.

The survey also revealed strong support for BCI technology, which enables direct communication between the brain and external systems. Respondents expressed high comfort and enthusiasm for using BCI to control environmental factors such as lighting and temperature. The concept of interacting with residential spaces through cognitive activity was particularly well received, reflecting significant interest in this novel approach. These findings align with existing research, emphasizing the transformative potential of integrating BIM and BCI in residential environments.

Research by Kuleto et al. underscores the synergies between BIM and BCI in creating sustainable neuro-responsive habitats that prioritize user well-being and environmental preservation. Yang et al. highlight the underexplored applications of BCIs in architectural design, advocating for customized solutions that enhance the design process. Similarly, Zhang et al. call for a shift towards sustainable and environmentally responsible urban development.

Despite these technologies' promising potential, the study also identifies challenges that need to be addressed, including precision, privacy, and scalability issues. Ethical considerations such as data security and informed user consent are critical, as emphasized by Sun et al., who recommend thorough reviews better to understand the psychophysiological responses of individuals within architectural settings.

Several technological barriers impede the widespread adoption of integrated BCI and BIM systems. Key challenges include data interoperability, system scalability, and the accurate interpretation of brain feedback. However, the study provides a roadmap for future advancements, suggesting that ongoing technological improvements and interdisciplinary collaboration are necessary to overcome these hurdles. The research also indicates the potential for expanding these technologies into larger urban development projects, paving the way for reimagining smart cities through a sustainable framework.

The integration of BIM and BCI offers a groundbreaking approach to architectural design and urban planning, fostering environments that are not only sustainable but also highly attuned to their inhabitants' mental and emotional states. While technological and ethical challenges remain, the path forward is marked by the potential for significant progress in creating human-centered, ecologically sustainable, and adaptable living spaces.

Merging BIM and BCI technologies could lead to breakthroughs in creating intelligent, sustainable, and responsive residential environments. These advancements can enhance the quality of life while aligning with broader sustainability goals, such as reducing energy consumption and minimizing environmental impact. Achieving the full potential of these transformative technologies requires ongoing research, development, and careful attention to ethical considerations.

Table 3 details the research survey's design process, which aimed to understand the integration of BCI and BIM technologies in residential settings. The table provides a comprehensive overview of the methodological steps to ensure the survey's validity and reliability. It outlines each process stage's purpose, key actions, and outcomes. The development process began with a review of relevant literature and consultations with subject matter experts, shaping the survey questionnaire's content.

Subsequent steps included pilot testing, distribution of the final survey, and analysis of the collected data. Ethical considerations were meticulously addressed to protect participant confidentiality and ensure data integrity. This careful approach resulted in a reliable and valid survey instrument, as demonstrated by the table, providing a robust basis for analyzing the potential adoption and impact of BCI and BIM technologies in smart home environments. This thorough process ensured the gathered data were relevant and reliable, forming a solid foundation for exploring the future of smart, adaptive residential spaces.

Table 3. Comprehensive overview of the survey creation process.

Stage	Description	Key Activities	Outcomes
Literature Review and Expert Consultation	A comprehensive review of the literature on BCI, BIM, and smart home technologies is combined with discussions with authorities in related domains.	Interview of specialists and review of information from the industry and scholarly journals.	Essential concepts, topics, and possible survey questions were uncovered.
Questionnaire Development	A structured questionnaire was developed using literature and expert input information. It consisted of closed-ended questions using a Likert scale and open-ended ones.	The process involves formulating inquiries, selecting suitable measurement scales, and creating the initial survey.	A pilot-testable survey draft with relevant and unambiguous questions.
Pilot Testing	Pilot testing with a small group to improve question clarity, relevance, and comprehensibility for uniform interpretation.	Testing of the survey with a limited group, listening to their comments, and adjusting the questions.	Question clarity and relevancy were enhanced in the revised survey instrument.
Final Survey Administration	The final survey was sent out to more people using an internet platform; 130 people were included in the sample.	Sending out of invites and updating response rates using an online survey tool.	A total of 130 valid replies have been completed for analysis.
Sampling Method	Participants with a foundational knowledge of BCI and BIM technologies were selected using a purposeful sampling strategy.	Conducting participant recruitment, establishing inclusion criteria, and guaranteeing a statistically valid sample.	A representative cross-section of the population with sufficient knowledge to offer helpful feedback on the issues at hand.
Response Rate	Received and processed all 130 questions, ensuring a perfect response rate.	Keeping in touch with participants to make sure they are actively engaged and that their responses are complete.	The data used for analysis will be vital if the response rate is high.
Quantitative Analysis	Using descriptive statistics, we can determine where people are leaning and how spread out they are by examining the results of Likert-scale questions.	Creation of descriptive statistics from survey data using statistical software.	Comprehensive statistical analysis that sheds light on the perspectives and attitudes of the participants.
Qualitative Analysis	Supplementing the quantitative data with thematic analysis of open-ended replies helps to discover common themes and ideas, providing a broader picture.	Organizing free-form replies into categories, coding them, and looking for patterns.	Comprehensive analysis based on in-depth qualitative insights supplemented by quantitative results.
Ethical Considerations	Verification that the survey adheres to ethical guidelines on data protection, informed consent, and secrecy.	Getting consent forms ready, making replies anonymous, and keeping data secure.	Adhering to ethical standards protects participants' confidence in the study process and its honesty.
Data Integration and Interpretation	To conduct a thorough analysis, by combining quantitative and qualitative data.	Creation of the analysis report by comparing quantitative findings with qualitative observations.	A comprehensive evaluation that integrates numerical data with detailed contextual knowledge.
Reporting and Dissemination	Final report preparation, data visualization, and stakeholder dissemination.	Report writing, graphic design, and presenting results to appropriate audiences.	Detailed report outlining the survey's methodology, results, and potential future study directions.

5.5. Summary of Key Analytical Focus Areas

The Results and Discussion section provides a detailed analysis of the data collected from the Likert-scale survey, highlighting vital statistical outcomes and trends that reflect the participants' viewpoints. The survey used a scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), enabling a precise measurement of participant sentiments toward specific statements. This method allowed researchers to effectively quantify the levels of agreement or disagreement among participants.

A notable finding from the research is the strong trend toward agreement, as most participants selected "4" on the scale, indicating a high level of alignment with the examined statements. This suggests a consensus among respondents regarding the statements presented in the survey. The statistical measures, including a mean of 4.05, a median of 4, and a mode of 4, further reinforce this pattern of positive consensus. These values indicate that most responses clustered around a similar agreement point, highlighting participants' shared perspectives.

The absence of solid dissent, such as the response of "1" (Strongly Disagree), is particularly significant. It suggests minimal resistance to the ideas or concepts being explored within the survey. This lack of opposition implies a broadly favorable reception among the surveyed group, making it easier to draw general conclusions about the overall attitude toward the subject matter.

The distribution of responses, visualized through figures and charts, showcases the uniformity of opinions and provides a foundation for assessing the broader implications of these findings. This reaction consistency is crucial for understanding the potential acceptance or integration of the studied technologies or concepts.

This section discusses the positive trends observed and addresses the survey's limitations. Acknowledging these shortcomings is essential for understanding the scope of the data and ensuring that the conclusions drawn are based on realistic interpretations. This part of the discussion also suggests directions for future research, offering opportunities to explore areas that may not have been fully addressed in the current study.

The analysis of the survey results emphasizes the importance of the collected data in shaping the study's conclusions. It establishes a framework for the subsequent discussion, which focuses on aligning these findings with the broader objectives of the research. This alignment is critical to understanding how the participants' perspectives support or challenge the theoretical and practical implications presented in the earlier sections of the study.

6. Conclusions

The study comprehensively analyzes attitudes and perceptions toward advanced technologies in residential settings, particularly Building Information Modeling (BIM) and the Brain–Computer Interface (BCI). It concludes with vital theoretical contributions and practical implications, identifying fundamental limitations and suggesting directions for future research.

The study advances the concept of neuro-responsive habitats, where cognitive input from BCIs enables real-time, adaptive changes in the built environment. This approach contributes significantly to user-centered, intelligent architectural design, and sustainable building discourse. By integrating human cognitive and emotional data into architectural processes, the study connects modern technology with architecture, positioning the built environment as an active participant in enhancing occupant well-being. Integrating BCI and BIM within the Construction 5.0 framework deepens the theoretical understanding of optimizing human–environment interactions for sustainability. It highlights how neuro-responsive habitats synchronize occupants' mental and physiological conditions with adaptive environmental controls.

The study offers architects, engineers, and construction professionals valuable insights about implementing BCI–BIM integration in smart home environments. The technology's potential to create adaptable, occupant-responsive environments that improve comfort

and energy efficiency marks a significant advancement in sustainable design. The findings provide practical strategies for integrating these technologies, showcasing how BCI-driven adjustments to environmental parameters like lighting, temperature, and spatial arrangements can enhance residents' quality of life while minimizing ecological impact. Additionally, the research demonstrates how combining BCI and BIM could influence future industry practices, fostering greater creativity in sustainable building and smart home design.

The study acknowledges several study limitations. A major challenge is the scalability of the BCI-BIM architecture. While the integration shows promise in controlled environments, applying it to larger residential complexes and urban areas remains challenging. Another issue is the accuracy of cognitive data obtained through BCI technology, which may include discrepancies that affect the system's reliability in adjusting settings based on real-time input. Concerns about data privacy and security, especially when handling sensitive neurological information, highlight the need for improved data-handling practices that prioritize security and trust.

To address these limitations, future studies should explore advanced statistical methods for evaluating the BCI-BIM framework across diverse demographics and settings. Comparative analyses with other intelligent building technologies, such as AI-driven systems and Internet of Things (IoT) applications, could provide a more comprehensive understanding of how these technologies complement each other. Expanding the research to include urban-scale initiatives would offer insights into how neuro-responsive settings function in smart cities and more significant residential developments.

Further research should also focus on enhancing the precision of BCI technology, particularly in interpreting cognitive data accurately. This improvement is essential for increasing the system's effectiveness. Additionally, ongoing scrutiny of ethical concerns related to data privacy, security, and user consent is crucial. Implementing robust data protection measures will ensure BCI-driven systems comply with ethical standards, especially in managing sensitive cognitive data.

The study emphasizes the importance of multidisciplinary collaboration among architects, technologists, ethicists, and legal experts to establish regulatory frameworks that guide the ethical application of BCI-BIM technology. Such collaboration is critical for addressing the ethical challenges identified and ensuring these technologies' responsible development and use.

This study establishes a strong foundation for integrating BCI and BIM within the context of Construction 5.0, offering both theoretical insights and practical recommendations. Addressing the identified challenges and expanding the scope of future research will be crucial for fully realizing the transformative potential of neuro-responsive, sustainable ecosystems. The ideas explored in this study illustrate the potential for advanced technologies to revolutionize residential environments by making them more adaptive, sustainable, and user-focused, thus shaping the future of intelligent living spaces.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/su16219393/s1>.

Author Contributions: Conceptualization, A.A. (Amjad Almusaed), I.Y., J.A.M. and A.A. (Asaad Almssad); methodology, A.A. (Amjad Almusaed), I.Y., J.A.M. and A.A. (Asaad Almssad); validation, A.A. (Amjad Almusaed), I.Y. and J.A.M.; formal analysis, A.A. (Amjad Almusaed), A.A. (Asaad Almssad), J.A.M.; investigation, A.A. (Amjad Almusaed), I.Y., J.A.M. and A.A. (Asaad Almssad); resources, A.A. (Amjad Almusaed), J.A.M. and A.A. (Asaad Almssad); data curation, A.A. (Amjad Almusaed), J.A.M. and A.A. (Asaad Almssad); writing—original draft preparation, A.A. (Amjad Almusaed), I.Y. and A.A. (Asaad Almssad); writing—review and editing, A.A. (Amjad Almusaed), I.Y. and A.A. (Asaad Almssad); visualization, A.A. (Amjad Almusaed), J.A.M. and A.A. (Asaad Almssad); supervision, A.A. (Amjad Almusaed), I.Y., J.A.M. and A.A. (Asaad Almssad); project administration, A.A. (Amjad Almusaed), I.Y. and A.A. (Asaad Almssad). All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The Institutional Review Board (IRB) clearance was considered unnecessary for this study since it was classified as a minimal-risk research. The study's approach, utilizing anonymous Likert-scale surveys, presented exceedingly little danger to participants. Furthermore, no personally identifying information was gathered, guaranteeing total anonymity during the study. Research operations that preserve anonymity and entail low risk are free from IRB assessment under Swedish institutional policies and legislation by national ethical standards and guidelines. Notwithstanding the exception, the researchers meticulously complied with ethical norms to maintain the study's integrity. Thorough methods were implemented to manage and analyze anonymous data, rigorously preserving participant anonymity and privacy. These actions were enacted by recognized ethical frameworks to maintain the highest levels of research ethics, notwithstanding the lack of direct IRB monitoring. The authors ensured that the study adhered to all requisite legal and institutional regulations, safeguarding participants' rights and preserving the study's ethical integrity.

Informed Consent Statement: Before the study began, participants had to answer a Likert-scale questionnaire after undergoing a computerized informed consent procedure. This method was carefully crafted to adhere to the highest ethical standards and to foster transparency throughout the research. Participants were provided with an extensive briefing on the study's aims, objectives, and survey techniques to ensure a thorough understanding of their participation. They were adequately apprised of their rights, encompassing the voluntary character of their engagement and their unequivocal right to exit at any moment without repercussions. The questionnaire's structured framework, with predetermined response categories, was explicitly conveyed to participants to elucidate the use of their replies in the study's analysis and results. This digital consent protocol was established to rigorously adhere to ethical standards, guaranteeing the protection of participants' privacy, anonymity, and the confidentiality of their replies. The information disseminated to participants was as follows: To ensure participants gained a comprehensive understanding of the study's scope, the research commenced with a detailed overview of its aims and objectives, clearly articulating the purpose and significance of the investigation. Participants' roles were meticulously outlined, and a Likert-scale questionnaire was administered as a core component of the data collection process. Emphasizing ethical standards, particularly regarding participant autonomy and trust, it was underscored that participation was entirely voluntary, allowing individuals to withdraw at any time without repercussions. The study demonstrated a strong commitment to confidentiality and privacy by implementing robust protocols for anonymous data collection and handling via the Likert-scale questionnaire. Throughout the research process, participants' rights and ethical principles were rigorously upheld, with this structured and transparent approach ensuring that all involved were thoroughly informed and protected.

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

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